

Measuring Causes and Consequences of Errors in RNA

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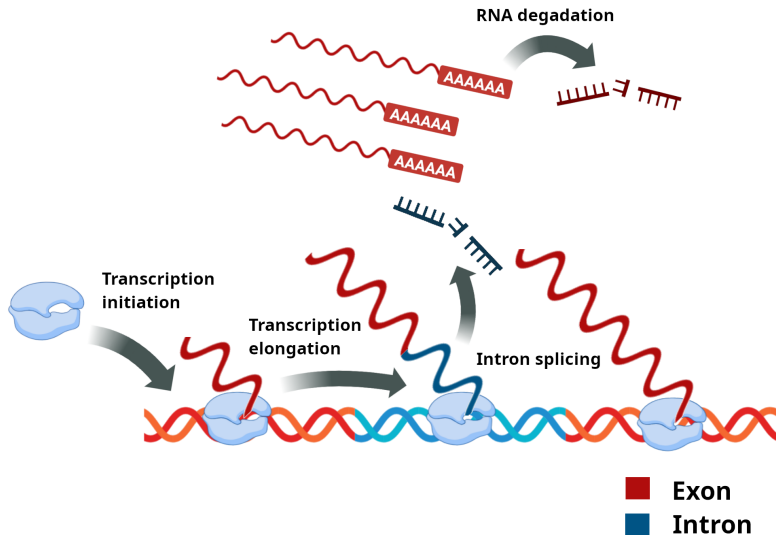
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The first part turned into a more complex modeling of the RNA metabolism, and I focused almost entirely on it during the last year.

RNA metabolism overview

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The dynamics may be described by an ODE system:

$$\begin{aligned}\dot{T}_{\text{nascent}} &= k_{\text{init}} - k_{\text{elong}} \cdot T_{\text{nascent}} \cdot L_{\text{gene}}^{-1} \\ \dot{I}_{\text{nascent}} &= k_{\text{init}} - k_{\text{elong}} \cdot I_{\text{nascent}} \cdot L_{\text{intron}}^{-1} \\ \dot{T}_{\text{mature}} &= k_{\text{elong}} \cdot T_{\text{nascent}} \cdot L_{\text{gene}}^{-1} - k_{\text{deg}} \cdot T_{\text{mature}} \\ \dot{I}_{\text{unspliced}} &= k_{\text{elong}} \cdot I_{\text{nascent}} \cdot L_{\text{intron}}^{-1} - k_{\text{splice}} \cdot I_{\text{unspliced}}\end{aligned}$$

where we have

- Kinetic rates: $[k_{\text{init}}] = \text{polymerases} \cdot \text{s}^{-1}$, $[k_{\text{elong}}] = \text{bp} \cdot \text{s}^{-1}$,
 $[k_{\text{splice}}] = \text{s}^{-1}$, $[k_{\text{deg}}] = \text{s}^{-1}$
- Gene and intron lengths: $[L_{\text{gene}}] = [L_{\text{intron}}] = \text{bp}$

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We may solve the ODE system for steady state solution:

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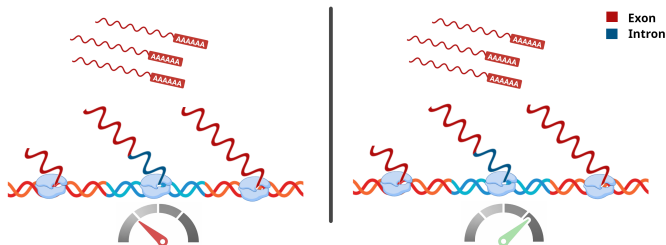
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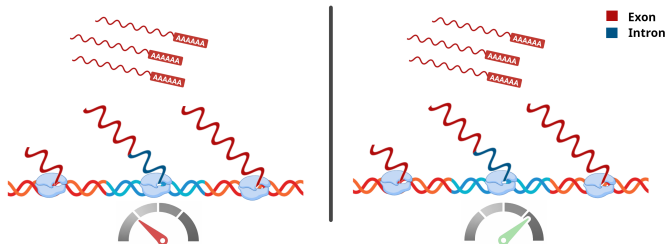
- The steady state quantities depends only on the ratios of kinetic rates.
- Therefore, scaling all kinetic rate by some constant $c \in \mathbb{R}_+$ does not change the steady state.

Non-identifiability of kinetic rates from steady state

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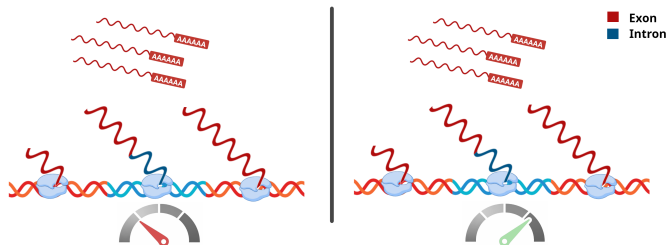


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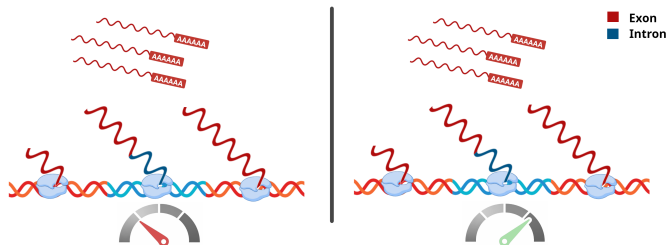
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- When comparing samples (treatment vs. control), only differences of LFC can be estimated and statistically tested.
- E.g., we may estimate the value of

$$\text{LFC}(k_{\text{elong}}) - \text{LFC}(k_{\text{deg}})$$

and test the hypothesis

$$H_0 : \text{LFC}(k_{\text{elong}}) - \text{LFC}(k_{\text{deg}}) = 0$$

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- Therefore, steady-state data cannot distinguish $f(\mathbf{x})$ from $c \cdot f(\mathbf{x})$. Consequently, only ratios of rates are identifiable.
- This type of structural non-identifiability is common in RNA metabolism models.

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 - E.g., inhibiting/releasing transcription, and/or using labeled nucleotides.

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- With RNA-seq, we observe the number of reads r_i corresponding to the molecules of interest, which scales as

$$r_i \propto \frac{|A_i|}{|S_i|} = \frac{|A_i|}{|A_i| + |A_i^c|}$$

where A_i^c is a complement of A_i in S_i .

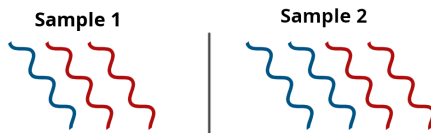
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- In both cases, the fraction of intronic reads is identical, making the two scenarios indistinguishable from RNA-seq data.

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- To overcome these issues, we need either additional measurements, and/or different types of data than total RNA-seq.

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Thank you for your attention!